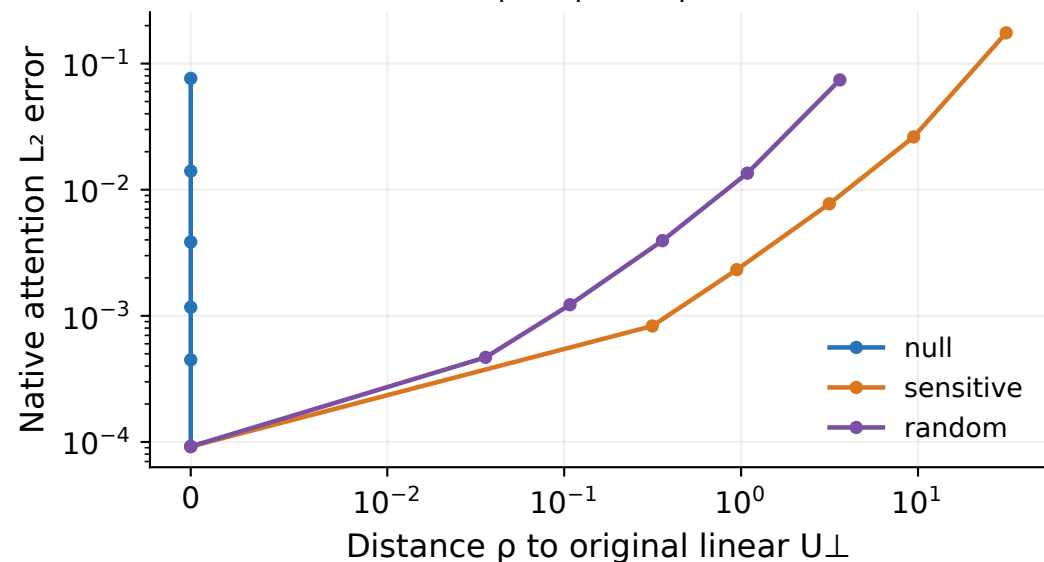
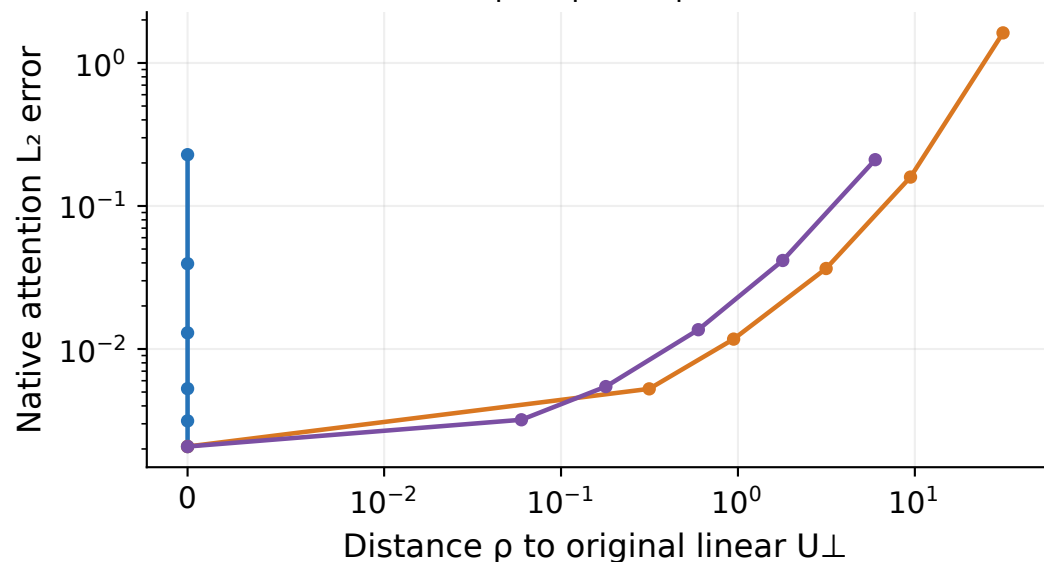


Fixed optimized r: equal-norm query perturbations

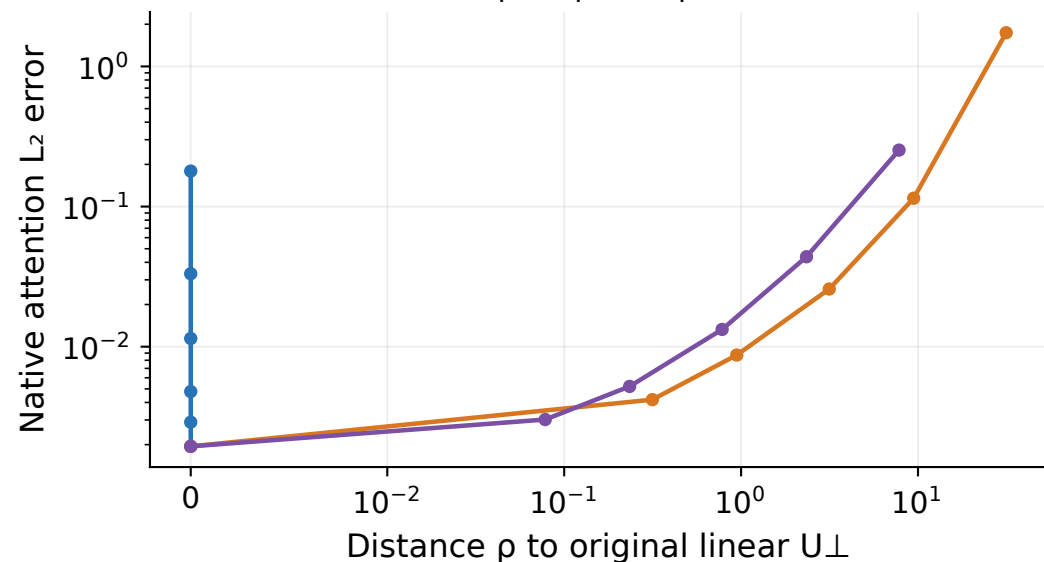
1 prompt, 1 input



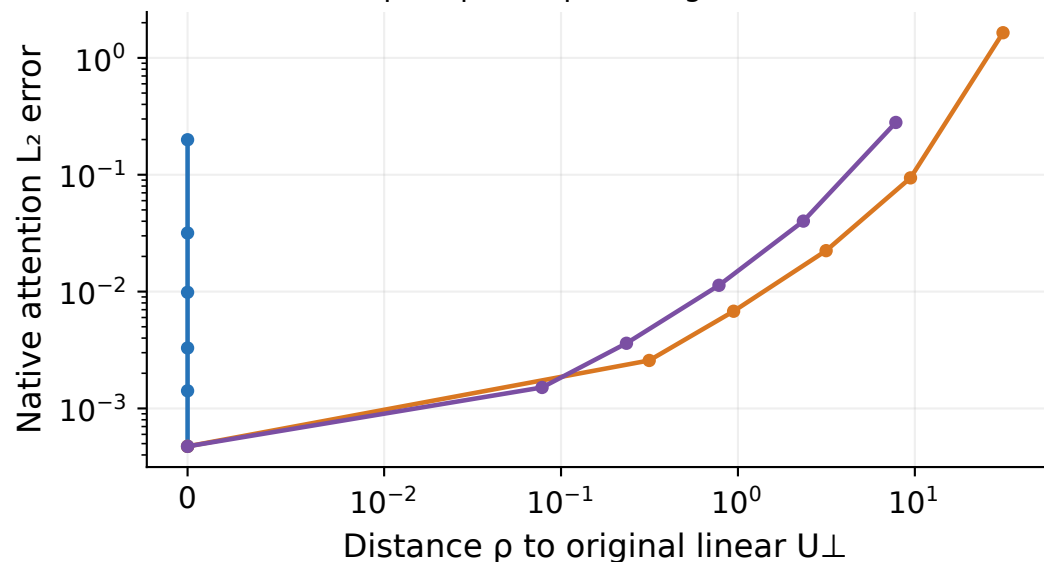
4 prompt, 1 input



4 prompt, 4 input



4 prompt, 1 input + 3 generated



$U_{\perp}$ is a linear-theory diagnostic. Native QK normalization / RoPE need not preserve matching, even at $p = 0$.